In the Name of God

Introduction to Machine Learning (25737-2)

Problem Set 3

Spring Semester 1403-04

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Due on Farvardin 23, 1404 at 23:59

1 Logistic Regression

We consider here a discriminative approach for solving the classification problem illustrated in Figure 1

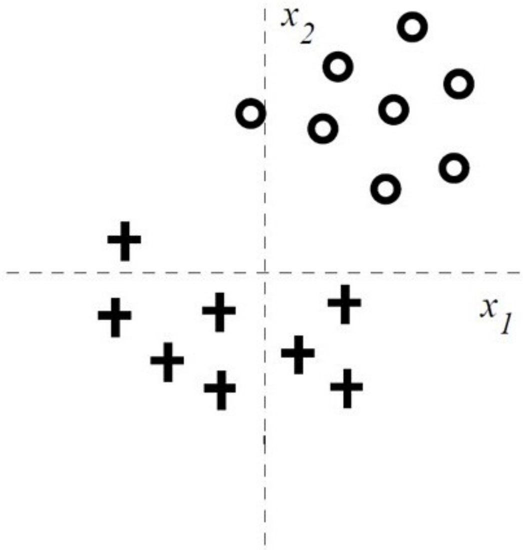

Figure 1: The 2-dimensional labeled training set, where '+' corresponds to class $y = 1$ and 'O' corresponds to class $y = 0$.

1. We attempt to solve the binary classification task depicted in Figure 1 with the simple linear logistic regression model

$$P(y = 1|\vec{x}, \vec{w}) = g(w_0 + w_1x_1 + w_2x_2) = \frac{1}{1 + \exp(-w_0 - w_1x_1 - w_2x_2)}.$$

Notice that the training data can be separated with *zero* training error with a linear separator.

Consider training *regularized* linear logistic regression models where we try to maximize

$$\sum_{i=1}^{n} \log\left(P(y_i|x_i, w_0, w_1, w_2)\right) - Cw_j^2$$

for very large C. The regularization penalties used in penalized conditional log-likelihood estimation are $-Cw_j^2$, where $j = \{0, 1, 2\}$. In other words, only one of the parameters is regularized in each case.

Given the training data in Figure 1, how does the training error change with regularization of each parameter w_j? State whether the training error increases or stays the same (zero) for each w_j for very large C. Provide a brief justification for each of your answers.

(a) By regularizing w_2

(b) By regularizing w_1

(c) By regularizing w_0

2. If we change the form of regularization to L1-norm (absolute value) and regularize w_1 and w_2 only (but not w_0), we get the following penalized log-likelihood

$$\sum_{i=1}^{n} \log P(y_i|x_i, w_0, w_1, w_2) - C(|w_1| + |w_2|).$$

Consider again the problem in Figure 1 and the same linear logistic regression model

$$P(y = 1|\vec{x}, \vec{w}) = g(w_0 + w_1 x_1 + w_2 x_2).$$

(a) As we increase the regularization parameter C which of the following scenarios do you expect to observe? (Choose only one) Briefly explain your choice:

( ) First w_1 will become 0, then w_2.

( ) First w_2 will become 0, then w_1.

( ) w_1 and w_2 will become zero simultaneously.

( ) None of the weights will become exactly zero, only smaller as C increases.

(b) For very large C, with the same L1-norm regularization for w_1 and w_2 as above, which value(s) do you expect w_0 to take? Explain briefly. (Note that the number of points from each class is the same.) (You can give a range of values for w_0 if you deem necessary).

(c) Assume that we obtain more data points from the '+' class that corresponds to $y = 1$ so that the class labels become unbalanced. Again for very large C, with the same L1-norm regularization for w_1 and w_2 as above, which value(s) do you expect w_0 to take? Explain briefly. (You can give a range of values for w_0 if you deem necessary).

2 Kernel Regression

Let's consider the non-parametric kernel regression setting. In this problem, you will investigate the locally univariate linear regression where the estimator is of the form:

$$\hat{f}(x) = \beta_1 + \beta_2 x$$

and the solution for parameter vector $\beta = [\beta_1\ \beta_2]$ is obtained by minimizing the weighted least square error:

$$J(\beta_1, \beta_2) = \sum_{i=1}^{n} W_i(x)(Y_i - \beta_1 - \beta_2 X_i)^2 \quad \text{where} \quad W_i(x) = \frac{K\left(\frac{X_i - x}{h}\right)}{\sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right)},$$

where K is a kernel with bandwidth h. Observe that the weighted least squares error can be expressed in matrix form as

$$J(\beta_1, \beta_2) = (Y - A\beta)^T W (Y - A\beta),$$

where Y is a vector of n labels in the training example, W is an $n \times n$ diagonal matrix with weight of each training example on the diagonal, and

$$A = \begin{bmatrix} 1 & X_1 \\ 1 & X_2 \\ \vdots & \vdots \\ 1 & X_n \end{bmatrix}$$

1. Derive an expression in matrix form for the solution vector $\hat{\beta}$ that minimizes the weighted least square.
2. When is the above solution unique?
3. If the solution is not unique, one approach is to optimize the objective function J using gradient descent. Write the update equation for gradient descent in this case.
Note: Your answer must be expressed in terms of the matrices defined above.
4. Can you identify the signal plus noise model under which maximizing the likelihood (MLE) corresponds to the weighted least squares formulation mentioned above?
5. Why is the above setting non-parametric? Mention one advantage and one disadvantage of nonparametric techniques over parametric techniques.

3 Bayes Optimal Classification

In classification, the loss function we usually want to minimize is the 0/1 loss:

$$\ell(f(x), y) = \mathbf{1}\{f(x) \neq y\}$$

where $f(x), y \in \{0, 1\}$ (i.e., binary classification). In this problem, we will consider the effect of using an asymmetric loss function:

$$\ell_{\alpha,\beta}(f(x), y) = \alpha \mathbf{1}\{f(x) = 1, y = 0\} + \beta \mathbf{1}\{f(x) = 0, y = 1\}$$

Under this loss function, the two types of errors receive different weights, determined by $\alpha, \beta > 0$.

1. Determine the Bayes optimal classifier, i.e., the classifier that achieves minimum risk assuming $P(x, y)$ is known, for the loss $\ell_{\alpha,\beta}$ where $\alpha, \beta > 0$.
2. Suppose that the class $y = 0$ is extremely uncommon (i.e., $P(y = 0)$ is small). This means that the classifier $f(x) = 1$ for all x will have good risk. We may try to put the two classes on even footing by considering the risk:
$$R = P(f(x) = 1 \mid y = 0) + P(f(x) = 0 \mid y = 1)$$
Show how this risk is equivalent to choosing a certain α, β and minimizing the risk where the loss function is $\ell_{\alpha,\beta}$.
3. Consider the following classification problem. I first choose the label $Y \sim \text{Bernoulli}(\frac{1}{2})$, which is 1 with probability $\frac{1}{2}$. If $Y = 1$, then $X \sim \text{Bernoulli}(p)$; otherwise, $X \sim \text{Bernoulli}(q)$. Assume that $p > q$. What is the Bayes optimal classifier, and what is its risk?
4. Now consider the regular 0/1 loss ℓ, and assume that $P(y = 0) = P(y = 1) = \frac{1}{2}$. Also, assume that the class-conditional densities are Gaussian with mean μ_0 and co-variance Σ_0 under class 0, and mean μ_1 and co-variance Σ_1 under class 1. Further, assume that $\mu_0 = \mu_1$.

For the following case, draw contours of the level sets of the class conditional densities and label them with $p(x|y = 0)$ and $p(x|y = 1)$. Also, draw the decision boundaries obtained using the Bayes optimal classifier in each case and indicate the regions where the classifier will predict class 0 and where it will predict class 1.

$$\Sigma_0 = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}, \quad \Sigma_1 = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$$

4 * Logistic Regression for Sparse Data

In many real-world scenarios our data has millions of dimensions, but a given example has only hundreds of non-zero features. For example, in document analysis with word counts for features, our dictionary may have millions of words, but a given document has only hundreds of unique words. In this question we will make ℓ_2 regularized SGD efficient when our input data is sparse. Recall that in ℓ_2 regularized logistic regression, we want to maximize the following objective (**in this problem we have excluded** w_0 **for simplicity**):

$$F(\mathbf{w}) = \frac{1}{N}\sum_{j=1}^{N} l(\mathbf{x}^{(j)}, y^{(j)}, \mathbf{w}) - \frac{\lambda}{2}\sum_{i=1}^{d}\mathbf{w}_i^2$$

where $l(\mathbf{x}^{(j)}, y^{(j)}, \mathbf{w})$ is the logistic objective function

$$l(\mathbf{x}^{(j)}, y^{(j)}, \mathbf{w}) = y^{(j)}\left(\sum_{i=1}^{d}\mathbf{w}_i x_i^{(j)}\right) - \ln\left(1 + \exp\left(\sum_{i=1}^{d}\mathbf{w}_i x_i^{(j)}\right)\right)$$

and the remaining sum is our regularization penalty.
When we do stochastic gradient descent on point $(\mathbf{x}^{(j)}, y^{(j)})$, we are approximating the objective function as

$$F(\mathbf{w}) \approx l(\mathbf{x}^{(j)}, y^{(j)}, \mathbf{w}) - \frac{\lambda}{2}\sum_{i=1}^{d}\mathbf{w}_i^2$$

Definition of sparsity: Assume that our input data has d features, i.e. $\mathbf{x}^{(j)} \in \mathbb{R}^d$. In this problem, we will consider the scenario where $\mathbf{x}^{(j)}$ is sparse. Formally, let s be the average number of nonzero elements in each example. We say the data is sparse when $s \ll d$. In the following questions, **your answer should take the sparsity of $\mathbf{x}^{(j)}$ into consideration when possible.**

Note: When we use a sparse data structure, we can iterate over the non-zero elements in $O(s)$ time, whereas a dense data structure requires $O(d)$ time.

1. Let us first consider the case when $\lambda = 0$. Write down the SGD update rule for $\mathbf{w}_i$ when $\lambda = 0$, using step size η, given the example $(\mathbf{x}^{(j)}, y^{(j)})$.

2. If we use a dense data structure, what is the average time complexity to update $\mathbf{w}_i$ when $\lambda = 0$? What if we use a sparse data structure? Justify your answer in one or two sentences.

3. Now let us consider the general case when $\lambda > 0$. Write down the SGD update rule for $\mathbf{w}_i$ when $\lambda > 0$, using step size η, given the example $(\mathbf{x}^{(j)}, y^{(j)})$.

4. If we use a dense data structure, what is the average time complexity to update $\mathbf{w}_i$ when $\lambda > 0$?

5. Let $\mathbf{w}_i^{(t)}$ be the weight vector after t-th update. Now imagine that we perform k SGD updates on $\mathbf{w}$ using examples $(\mathbf{x}^{(t+1)}, y^{(t+1)}), \ldots, (\mathbf{x}^{(t+k)}, y^{(t+k)})$, where $x_i^{(j)} = 0$ for every example in the sequence. (i.e. the i-th feature is zero for all of the examples in the sequence). Express the new weight, $\mathbf{w}_i^{(t+k)}$ in terms of $\mathbf{w}_i^{(t)}, k, \eta$, and λ.

6. Using your answer in the previous part, come up with an efficient algorithm for regularized SGD when we use a sparse data structure. What is the average time complexity per example? (Hint: when do you need to update $\mathbf{w}_i$?)

5 * Linear Regression

(a) We are given a set of two dimensional inputs and their corresponding output pair: $\{x_{i,1}, x_{i,2}, y_i\}$. We would like to use the following regression model to predict y:

$$y_i = w_1^2 x_{i,1} + w_2^2 x_{i,2}$$

Derive the optimal value for w_1 when using least squares as the target minimization function (w_2 may appear in your resulting equation). Note that there may be more than one possible value for w_1.

(b) Now assume we only observe a single input for each output (that is, a set of $\{x, y\}$ pairs). We would like to compare the following two models on our input dataset (for each one we split into training and testing set to evaluate the learned model). Assume we have an unlimited amount of data:

$$\text{A: } y = w^2 x \qquad \text{B: } y = wx$$

Which of the following is correct (choose the answer that best describes the outcome):

1. There are datasets for which A would perform *better* than B
2. There are datasets for which B would perform *better* than A
3. Both 1 and 2 are correct.
4. They would perform equally well on all datasets.

6 * Lasso Regression

One of the regularization methods in linear regression problems is the Lasso method. In this method, L1 norm of the model's weights is included in the loss function. This causes the final solution of the problem to become more sparse. In this problem, we will see how the L1 norm term results in more sparsity.

Let $\mathbf{X} \in \mathbb{R}^{n \times d}$ be a matrix where each row is an observation with d features and we have a total of n observations. $\mathbf{y} \in \mathbb{R}^n$ is our label vector. Assume that $\mathbf{w} \in \mathbb{R}^d$ is the weight vector of our regression model and $\mathbf{w}^*$ is the optimum weight vector. Also assume that our data has been whitened, that is: $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$.

In Lasso regression the optimum weight vector is obtained as such:

$$\mathbf{w}^* = \arg\min_{\mathbf{w}} J_\lambda(\mathbf{w}),$$

where:

$$J_\lambda = \frac{1}{2}\|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \lambda\|\mathbf{w}\|_1.$$

1. First we show that whitening the dataset causes the features to be independent such that w_i^* can be concluded only from the ith feature. To prove this, first show that J_λ can be written as:

   $$J_\lambda(\mathbf{w}) = g(\mathbf{y}) + \sum_{i=1}^{d} f(\mathbf{X}_{:,i}, \mathbf{y}, w_i, \lambda),$$

   where $\mathbf{X}_{:,i}$ is the ith column of $\mathbf{X}$.

2. If $w_i \geq 0$, find w_i.

3. If $w_i < 0$, find w_i.

4. Based on previous sections, on what conditions w_i would equal to zero? How can this condition be applied?

5. As we know, in Ridge regression, regularization term in the loss function appears as $\frac{1}{2}\lambda\|\mathbf{w}\|_2^2$. In this case, when does w_i equal to zero? What is the difference between this case and the previous case?

7 Classification Algorithms

7.1 Naive Bayes

Recall that a Naive Bayes classifier assumes the features $x_1, x_2, \ldots$ are conditionally independent given the class label y, so that:

$$P(y|X = (x_1, \ldots, x_n)) \propto P(X, y) = P(X|y) \cdot P(y) = P(y) \cdot \prod_i P(x_i|y),$$

and consequently its prediction can be written as:

$$\arg\max_y P(y|X) = \arg\max_y P(y) \cdot \prod_i P(x_i|y). \tag{1}$$

Now, consider the following non-linear classifier with the classification rule of the following form:

$$\arg\max_y S\left(w_0 + \sum_{i=1}^{n} w_{y,i} x_i\right), \tag{2}$$

where S is the sigmoid function.

(a) Suppose the input features are binary i.e. $x_i \in \{0, 1\}$. Under this setting, rewrite the classification rule of the Naive Bayes classifier in Eqn. (1) to have the same form as that of the non-linear classifier in Eqn. (2) (**Hint:** Use the fact that X_i is binary).

(b) Express the weights $w_0, w_{y,i}$ (for $i = 1, \ldots, n$) in terms of the Naive Bayes probabilities by using $P(y)$ and $P(x_i|y)$. Assume that all Naive Bayes probabilities are non-zero.

7.2 Logistic Regression vs. Gaussian Naive Bayes

Now we no longer restrict Naive Bayes classifier to have boolean features. Consider the problem of classifying the following 2-dimensional datasets into two classes (represented by 'plus' and 'diamond') using two classifiers: Logistic Regression and Gaussian Naive Bayes. For the Gaussian Naive Bayes, assume that both classes have equal variances in the y-dimension.

For each of the three following datasets, **draw the Logistic Regression decision boundary with a solid line** and the **Gaussian Naive Bayes Boundary with a dashed line**. If any of the classifiers cannot classify the data correctly, write *one sentence* explaining why not on the right of the figure.

Note: You need to draw decision boundary only for the cases where the data can be classified correctly.